INCH-POUND

MIL-DTL-25446F
28 September 2022
SUPERSEDING
MIL-DTL-25446E
21 June 2012

DETAIL SPECIFICATION

TRANSMITTER, TEMPERATURE, ELECTRICAL RESISTANCE, ARMORED, BAYONET LOCKING TYPE, (-50 ºC TO 300 ºC)

Inactive for new design after 24 November 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

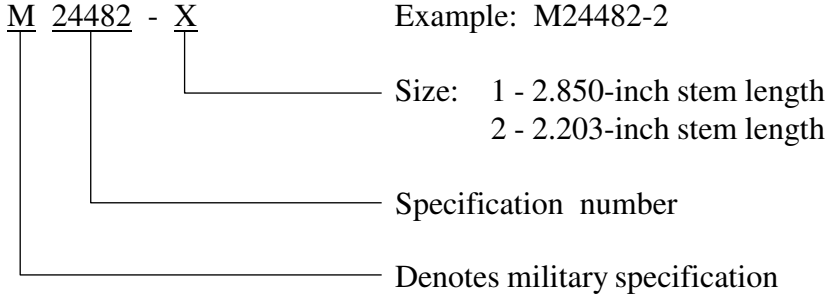
1. SCOPE

1.1 Scope. This specification covers the requirements for flexible sheath, electrical resistance-type, cylinder head temperature transmitters.

1.2 Classification. The temperature transmitters should be classified as specified below(see 6.2):

MS24482-1 - 2.850-inch stem length
MS24482-2 - 2.203-inch stem length

1.3 Part or identifying number (PIN). The PIN to be used for temperature transmitters acquired to this specification is created as follows. Legacy part numbers have been retained to support continued procurement:



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

QQ-B-654	- Brazing Alloys, Silver
AN5539	- Terminal - Thermocouple, Brass

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-1916	- DoD Preferred Methods Acceptance of Product
MIL-STD-31000	- Technical Data Packages
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	- Dissimilar Metals

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A581/A581M	- Standard Specification for Free-Machining Stainless Steel Wire and Wire Rods
ASTM E220	- Standard Test Method for Calibration of Thermocouples By Comparison Techniques
ASTM E230	- Standard Specification and Temperature-Electromotive Force (emf) Tables for Standardized Thermocouples

(Copies of these documents are available online at https://www.astm.org/.)

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SAE INTERNATIONAL

SAE-AMS-S-7720	- Steel, Corrosion-Resistant (18-8) Bars, Wire and Forging Stock (Aircraft Quality)

(Copies of this document are available online at https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Parts.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by MS or AN standard parts without alteration.
b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part number (see figure 1 and table I).

3.3 Materials. Materials that are not covered by applicable specifications shall be of the best commercial quality, of the lightest practicable weight, and entirely suitable for the purpose (see figure 1). Nonflammable materials shall be used in the construction of the transmitter.

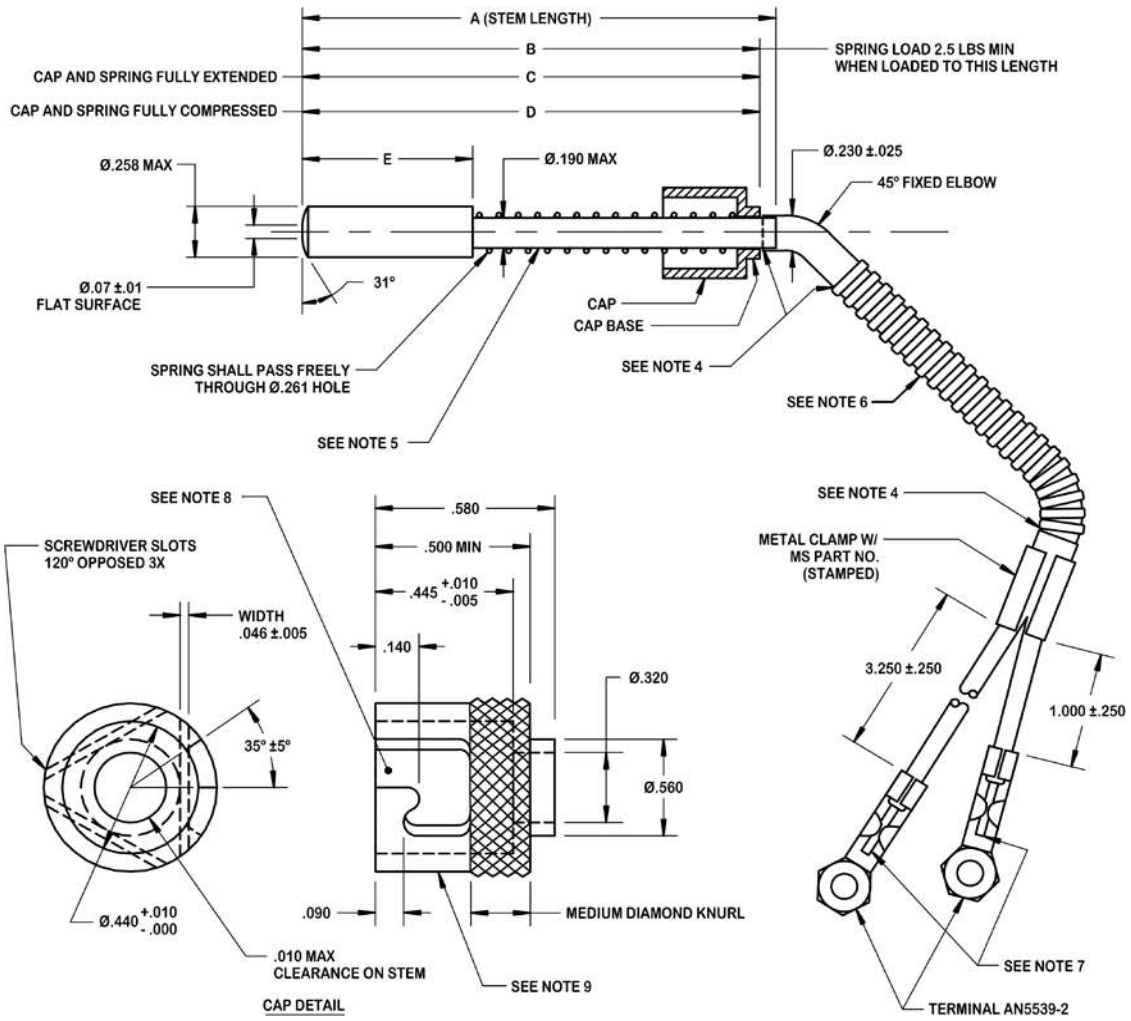
3.3.1 Corrosion resistance. Materials shall be of corrosion-resistant types or shall be suitably protected to resist corrosion.

3.3.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact. Dissimilar metals are defined in MIL-STD-889.

3.3.3 Toxic and corrosive fumes. The materials used in the construction of the temperature transmitter shall not release gases that combine with the atmosphere to form acidsor corrosive alkalis, nor shall they release toxic fumes when burned.

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3.3.4 Fungus-inert materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungus must be used, such materials shall be treated with a fungicidal agent, as approved by the procuring activity (see 6.2). Materials that are fungal nutrients and that are contained within a hermetically sealed enclosure need not be treated with a fungicide.



NOTES
1. All dimensions are in inches.
2. Unless otherwise specified, tolerances: 2 decimal places: ±0.02; 3 decimal places: ±0.010; angles: ±1°.
3. Overall length: 15.00 ±0.50 from tip of bulb to disconnect assembly on longest lead.
4. Silver braze: QQ-B-654, grade IV.
5. Steel stem: SAE AMS-S-7720, UNS S30200, or S31600.
6. Flexible conduit armor (shown) or braided sheath: Stainless steel, UNS S30200.
7. Silver braze wires to terminals using QQ-B-654, grade IV.
8. Slot to pass freely around Ø0.135 locking pin. Break sharp edges of slot.
9. Steel cap: SAE AMS-S-7720, UNS S30200 or S31600, or ASTM A581/A581M, UNS S41600.

FIGURE 1. Transmitter; temperature, electrical resistance, (-50 ºC to 300 ºC).

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TABLE I. Transmitter; temperature, electrical resistance, (-50 ºC to 300 ºC) - dimensions.

MS part no.	A	B	C Min	D Max	E Max
MS24482-1	2.850 ±0.030	2.50	2.62	1.78	1.00
MS24482-2	2.203 ±0.016	2.00	2.06	1.63	0.76

3.3.5 Protective treatment. When materials are used in the construction of the transmitters that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service use, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that is subject to cracking, chipping, or scaling with age or extremes of atmospheric conditions shall be avoided.

3.3.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design and construction. The temperature transmitter shall conform to the requirements contained herein and shall be constructed to withstand the normal strains or jars, vibrations, and such other conditions incident to shipping, storage, installation, and service without failure.

3.4.1 Temperature range. The temperature transmitter shall be designed to function satisfactorily throughout the temperature range -50 °C to 300 °C (-58 °F to 572 °F).

3.5 Performance. The temperature transmitter shall be capable of meeting the test requirements specified in section 4 when subjected to the following conditions:

a. Temperature: Temperatures ranging from -65 °C to 325 °C (-85 °F to 617 °F).
b. Insulation resistance: Satisfactory insulation resistance when a potential of 100 volts direct current (VDC) is applied between the connector terminals and any metal part of the transmitter.
c. Vibration: Vibration testing when performed in accordance with MIL-STD-810, section 514, category 8. F_i = 350 Hertz.
d. Humidity: Relative humidity up to 100 percent.
e. Fungus: Fungus growth as encountered in tropical climate.
f. Salt spray: Exposure to salt sea atmosphere.
g. Spring load: The applied spring load causing the distance between the temperature transmitter tip and the cap-base to be a distance of 2.5 inches for the MS24482-1, and 2.0 inches for the MS24482-2 shall be 2.5 pounds or greater.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-31000 shall govern changes in the manufacturer's part numbers.

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3.7 Insulation. The wires and insulation outside of the temperature transmitter stem shall be protected by a suitable insulation to resist deterioration from oil, engine cleaning fluids, and heat. The lead wires shall be constructed of flexible stranded wire. A stainless steel flexible sheath or conduit shall be used to protect the leads between the temperature transmitter stem and the metal clamp as indicated in figure 1.

3.8 Resistance element. The resistance element of the temperature transmitter shall be properly sealed within the temperature transmitter housing to withstand total immersion of the transmitter housing in a liquid bath.

3.9 Temperature-resistance characteristics. The temperature-resistance characteristics of the temperature transmitter shall conform to the values shown in table II.

TABLE II. Temperature-resistance characteristics for cylinder head transmitter.

Temperature °C	Resistance Ohms (Ω)	Tolerance Ω	Temperature °C	Resistance Ω	Tolerance Ω
-50	37.95	-0.60 / +1.80	130	89.20	±0.60
-40	40.15	-0.60 / +1.70	140	92.80	±0.60
-30	42.45	-0.60 / +1.60	150	96.50	±0.70
-20	44.85	-0.60 / +1.50	160	100.30	±0.70
-10	47.35	-0.60 / +1.40	170	104.20	±0.70
0	50.00	-0.50 / +1.35	180	108.20	±0.70
10	52.60	-0.50 / +1.33	190	112.30	±0.70
20	55.25	-0.50 / +1.30	200	116.50	±0.80
30	57.95	-0.50 / +1.27	210	120.80	±0.80
40	60.70	-0.50 / +1.24	220	125.20	±0.80
50	63.55	-0.60 / +1.20	230	129.70	±0.80
60	66.50	-0.60 / +1.17	240	134.30	±0.80
70	69.55	-0.60 / +1.17	250	139.00	±0.90
80	72.65	-0.60 / +1.15	260	143.80	±0.90
90	75.80	-0.60 / +1.15	270	148.75	±0.90
100	79.00	-0.60 / +1.10	280	153.85	±0.90
110	82.30	±0.60	290	159.10	±0.90
120	85.70	±0.60	300	164.50	±1.00

3.10 Electrical connection. The temperature transmitter shall be provided with electrical connection terminals as shown in figure 1 and specified in AN5539-2.

3.11 Weight. The weight of the complete temperature transmitter shall not exceed 0.20 pounds.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130 and shall include the PIN number (see 1.3) and the contractor's part number.

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3.13 Workmanship. The temperature transmitter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, impregnation, and brazing, and freedom of parts from burrs and sharp edges.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspections.

4.2.1 First article test samples. One sample shall be subjected to first article inspection.

4.2.2 First article tests. The first article tests shall consist of the inspections and tests specified in table III. The presence of any defect shall be the cause for rejection of the first article.

TABLE III. Inspections and tests.

Inspection or test	Individual tests	Sampling plan A tests	Sampling plan B tests	First Article tests	Test paragraph
Examination of product	X	X	X	X	4.5.1
Spring load test	X	X	X	X	4.5.2
Resistance check	X	X	X	X	4.5.3
Transmitter resistance test	-	X	X	X	4.5.4
High temperature exposure	-	X	X	X	4.5.5
Insulation resistance	-	X	X	X	4.5.6
Temperature gradient	-	-	X	X	4.5.7
Thermal response	-	-	X	X	4.5.8
Thermoelectric potential	-	-	X	X	4.5.9
Self-heating	-	-	X	X	4.5.10
Low temperature exposure	-	-	X	X	4.5.11
High temperature exposure of lead insulation	-	-	X	X	4.5.12
Vibration	-	-	-	X	4.5.13
Humidity	-	-	-	X	4.5.14
Fungus	-	-	-	X	4.5.15
Salt fog	-	-	-	X	4.5.16

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4.3 Conformance inspections. Conformance inspection shall be in accordance with MIL-STD-1916 and shall consist of individual tests (see 4.3.1) for each transmitter produced, as well as sampling A tests (see 4.3.2.1) or sampling plan B tests (see 4.3.2.2) as specified (see 6.2).

4.3.1 Individual tests. Each transmitter shall be subjected to the individual tests shown in table III.

4.3.2 Sampling plans and tests.

4.3.2.1 Sampling plan A. Sample units shall be subjected to the tests shown in table III for sampling plan A. Unless otherwise specified (see 6.2), the manufacturer, in accordance with MIL-STD-1916, shall select the type of sampling plan (attribute, variable, or continuous). The sample size shall be selected in accordance with verification level I of MIL-STD-1916.

4.3.2.2 Sampling plan B. Sample units shall be subjected to the tests shown in table III for sampling plan B. Unless otherwise specified (see 6.2), the manufacturer, in accordance with MIL-STD-1916, shall select the type of sampling plan (attribute, variable, or continuous). The sample size shall be selected in accordance with verification level I of MIL-STD-1916.

4.3.2.3 Test records. Test records must be available to substantiate the accumulated production quantities.

4.4 Test conditions.

4.4.1 Standard atmospheric conditions. Unless otherwise specified (see 6.2), all tests are to be conducted at atmospheric pressure (approximately 29.92 inches of mercury) and at room temperature (approximately 25 °C (77 °F)). When tests are performed at atmospheric pressure or room temperature that differ from these values, proper allowances shall be made for the difference from the specified conditions.

4.4.2 Test fixture. When specified, tests shall be conducted by using a test fixture in accordance with ASTM E220 with an axial temperature gradient of approximately 37.8 °C (100 °F) per inch when the temperature at the bottom of the well is 275 °C (527 °F). The temperature at the bottom of the well shall be determined by means of a master thermometer or thermocouple located in the fixture wall nearest to the bottom of the well. The temperature gradient of the fixture shall be determined by the use of an additional thermometer or thermocouple located 1 inch above the master thermometer/thermocouple.

4.4.3 Thermocouple electromotive force (emf). Corresponding values of temperature and emf for type J iron-constantan thermocouples are described in ASTM E230.

4.4.4 Test voltage. Unless otherwise specified, the temperature transmitter resistance shall be measured with an applied voltage of 1.5 ±0.2 VDC.

4.5.1 Examination of product. Each temperature transmitter shall be inspected to determine compliance with this specification with respect to materials, workmanship, and marking.

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4.5.2 Spring load. The tip of the temperature transmitter shall be fixed and the temperature transmitter shall be spring-loaded until its length from the tip of the temperature transmitter to the base of the case is 2.50 inches for the MS24482-1, and 2.00 inches for the MS24482-2 (see figure 1). The spring load shall be a minimum of 2.50 pounds.

4.5.3 Temperature transmitter resistance check. The temperature transmitter shall be immersed to the cap in an agitated liquid bath or other suitable non-gradient apparatus at a temperature of 200 ° ±0.1 °C (392 °F) for a period sufficient to insure that the transmitter has stabilized at the test temperature. The test voltage of 1.5 ±0.2 VDC shall be applied across the temperature transmitter terminals and the temperature transmitter resistance measured. The measured resistance shall be 116.50 ±0.80 Ω.

4.5.4 Temperature transmitter resistance test. For both first article and conformance testing, the temperature transmitter shall be immersed to the cap in an agitated liquid bath or other suitable non-gradient apparatus at the temperatures shown in 4.5.4.1 and 4.5.4.2 for a period sufficient to insure that the transmitter has stabilized ~~is~~ at the test temperature. The test voltage of 1.5 ±0.2 VDC shall be applied across the temperature transmitter terminals and the transmitter resistance measured. The measured resistance shall be within the tolerances specified in table II for the test temperature.

4.5.4.1 First article testing. The temperature transmitter shall be immersed at each of the 36 temperatures listed in table II for sufficient time to insure that the temperature transmitter has stabilized at the test temperature.

4.5.4.2 Conformance testing. For conformance testing, temperature transmitter resistance testing shall be performed at temperatures of 0 °C (32 °F), 100 °C (212 °F), and 300 °C (572 °F).

a. When the temperature transmitter resistance test is conducted as part of sampling plan A, tests shall be performed at temperatures of 0 °C (32 °F), 100 °C (212 °F), 200 °C (392 °F), and 300 °C (572 °F). The temperature tolerance for each measurement shall be ±0.1 °C (±0.18 °F).

b. When the temperature transmitter resistance test is conducted as part of sampling plan B, an additional test point of -50 ° ±0.1 °C (-58 ° ±0.18 °F) shall be included.

4.5.5 High temperature exposure. The temperature transmitter shall be inserted in a test fixture with a temperature at the bottom of the well of 325 ° ±5 °C (617 ° ±9 °F) for a period of 8 hours. The temperature at the bottom of the well shall then be reduced to a temperature of 275 ° ±5 °C (527 ° ±8 °F) and the temperature transmitter shall be subjected to this temperature for a period of 42 hours. The temperature transmitter shall then be withdrawn from the fixture and air-cooled to room temperature. The temperature transmitter shall then be subjected to the spring load test and the measured spring load shall be not less than 2.00 pounds. The temperature transmitter shall be subjected to the transmitter resistance test at a temperature of 275 °C (527 °F). The measured resistance shall be within the tolerance specified in table II for the test temperature. The test fixture used shall have an axial temperature gradient of 37.8 °C (100 °F). The well of the fixture shall be 7.0 inches minimum depth.

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4.5.6 Insulation resistance. A potential of 100 VDC shall be applied for 1 minute between each of the connector terminals and any other metal parts of the temperature transmitter assembly. The insulation resistance shall be not less than 5 megohms.

4.5.7 Temperature gradient. The temperature transmitter shall be inserted in the test fixture with a temperature at the bottom of the well of 200 ° ±1 °C (392 ° ±2 °F). The test voltage shall be applied across the temperature transmitter and the resistance measured. The measured resistance shall be within 1.7 Ω of the resistance value determined in the transmitter resistance test for the same temperature.

4.5.8 Thermal response. The temperature transmitter shall be inserted in the test fixture with a temperature at the bottom of the well of approximately 205 °C (400 °F). The transmitter shall remain in the fixture for a length of time sufficient to insure that the transmitter temperature has stabilized at the same temperature as the well temperature. The temperature transmitter shall then be removed from the fixture and air-cooled until the transmitter resistance for 200 °C (392 °F), as determined in the transmitter resistance test, is reached. The temperature transmitter shall then immediately be inserted in a second test fixture that has a temperature at the bottom of the well of 260 ° ±1 °C (500 ° ±2 °F). During the time required to properly insert the transmitter in the second fixture, the transmitter temperature will drop slightly below 200 °C. The transmitter temperature in the fixture shall be permitted to rise to 200 °C before response time is noted. The time required for the transmitter temperature to rise from 200 °C to 250 °C shall be noted. The resistance values for transmitter temperature determination at 200 °C and 250 °C shall be those values determined in the test fixture with temperatures, at the bottom of the well of 200 °C and 250 °C. The thermal response of the transmitter shall be considered as an average of three or more test runs and shall not exceed 20 seconds.

4.5.9 Thermoelectric potential. The temperature transmitter shall be inserted into the test fixture with a temperature at the bottom of the well of 200 ° ±5 °C (392 ° ±9 °F). The potentials produced in the transmitter winding shall not exceed one millivolt (mv).

4.5.10 Self-heating. The temperature transmitter shall be inserted into the fixture with a temperature at the bottom of the well of room temperature (approximately 25 °C (77 °F)). A current of 50 milliamperes (mA) shall be applied to the transmitter and the rise in resistance measured. The rise in resistance shall not exceed 3.0 Ω.

4.5.11 Low temperature exposure. The complete temperature transmitter assembly shall be subjected to a temperature of -65 ° ±2 °C (-85 ° ±3.6 °F) for a period of 48 hours. A continuity test shall be performed while at this low temperature. No open or short circuit shall occur and no other damage to the temperature transmitter shall result from this test.

4.5.12 High temperature exposure of lead insulation. The entire temperature transmitter assembly shall be subjected to a temperature of 175 ° ±2 °C (347 ° ±3.6 °F) for a period of 8 hours. There shall be no failure of the lead insulation, nor any other damage to the temperature transmitter assembly.

4.5.13 Vibration. A vibration test shall be performed in accordance with MIL-STD-810,

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method 514, category 8. The temperature transmitter shall be inserted in a test fixture with a temperature at the bottom of the well of 275 ° ±10 °C (527 ° ±16 °F) and the leads of the transmitter held by a stationary bracket in such a manner as to prevent pull on the leads during vibration. The leads shall be secured in such a manner that the angle between the longitudinal axis of the temperature transmitter and the transmitter leads is a minimum of 100 degrees. The test fixture used for this test shall be designed to approximate an axial temperature gradient of 37.8 °C. The well of the fixture shall be of the minimum depth (7.0 inches. Upon completion of the vibration test, the following tests, as specified herein, shall be conducted and the results shall be within the tolerance specified for the particular test:

a. Spring load: The spring load shall be a minimum of 2.0 pounds.
b. Transmitter resistance test.

4.5.14 Humidity. The temperature transmitter shall be subjected to humidity test, method 507, procedure I, of MIL-STD-810, except for a period of 5 cycles. There shall be no evidence of corrosion or other damage of the temperature transmitter. Following the humidity test, an insulation resistance test shall be conducted and the insulation resistance shall be not less than one megohm. Air drying of the temperature transmitter assembly prior to conducting the insulation resistance test is allowed.

4.5.15 Fungus. The temperature transmitter shall be subjected to a fungus test in accordance with method 508, procedure 1, of MIL-STD-810. There shall be no evidence of fungus growth on the temperature transmitter as a result of this test. Following the fungus test, an insulation resistance test shall be conducted and the insulation resistance shall be not less than one megohm.

4.5.16 Salt fog. The complete temperature transmitter assembly shall be subjected to a salt fog test in accordance with method 509 of MIL-STD-810 for a period of 50 hours. At the completion of this test, the transmitter may be cleaned externally with distilled water and air dried. Following the salt fog test, an insulation resistance test shall be conducted and the insulation resistance shall be not less than one megohm. The temperature transmitter shall then be subjected to the transmitter resistance tests. The temperature transmitter resistance shall not exceed the tolerances specified in table II. There shall be no evidence of corrosion or other damage as a result of the salt fog test.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The temperature transmitters covered by this specification are intended for use with an installed indicator in an aircraft to indicate the cylinder head temperature of air-cooled engine cylinders.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Classification (see 1.2).
c. First article, if required (see 3.1).
d. Fungicidal agents, if required (see 3.3.4).
e. Sampling plan A, as required (see 4.3.2.1).
f. Sampling plan B, as required (see 4.3.2.2).
g. Atmospheric test conditions, if different (see 4.4.1).
h. Packaging requirements (see 5.1).

6.3 Supersession. This specification supersedes MIL-T-25446D, dated 12 November 1986, and MS24482A, dated 22 April 1958.

6.4 Interoperability. The sensor, spring, and bayonet locking cap of the temperature transmitter are designed to connect to a corresponding female bayonet locking fitting in an aircraft engine block. Bayonet locking fittings for PIN's MS24482-1 and MS24482-2 should conform dimensionally to the dimensions shown in figure 1 and table I to allow secure connection to a fitting of the basic dimensions shown in figure 2.

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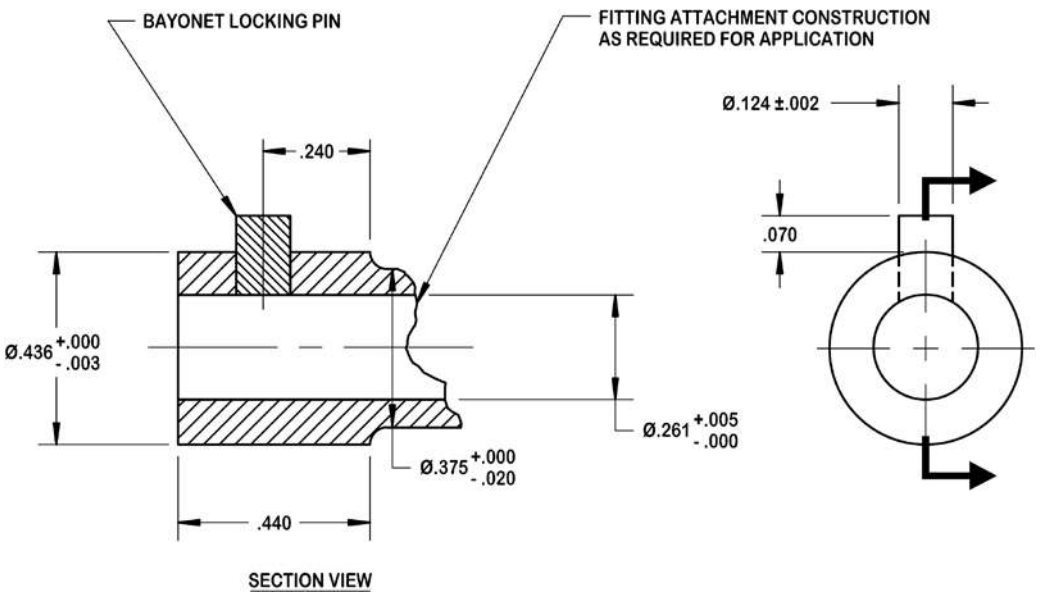


FIGURE 2. Female bayonet locking fitting - basic dimensions.

6.5 Subject term (key word) listing.

Electromotive
Emf
Iron constantan
Self heating
Sensor
Thermocouple
Thermosensor
Type J

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AV
Air Force - 70
DLA - GS

Preparing Activity:
DLA - GS1

(Project 6685-2022-001)

Review Activity:
Air Force - 99

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